Smaller direct quartic equations for ZFC consistency

Signed witnesses, sign normalization, and polynomial elimination

Research supplement prepared for Vladimir Reshetnikov

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Abstract

We construct an explicit integer-coefficient polynomial P of exact degree four in 3776 unrestricted integer variables. On the established ordinary mathematical dependencies of ProveIt's direct C222 construction, ZFC proves that P has an integer zero exactly when a finite syntactic ZFC refutation exists. The complete polynomial has 20194 nonzero monomials and coefficient height 47959. Its zero set is specified both by a fully expanded coefficient file and by 2015 quadratic residuals whose squares sum to P.

The starting direct quartic used 8,619 integers. The new reduction removes an unnecessary natural-number encoding of 221 signed quotients, normalizes 110 signs using exact polynomial symmetries, derives hundreds of other sign conditions from existing equations, and eliminates auxiliary coordinates with polynomial inverse maps. The resulting dimension reduction is 4843 variables, or 56.19%, with no increase in degree, while the coefficient height drops from 127,285,349 to 47,959. A second equation uses only 3,671 variables at the larger height 3,835,369,234. All new transformations have universal mathematical proofs and finite exact certificates checked by independent programs. This is a measured improvement of the direct, moderate-coefficient quartic family, not a minimum-variable record or a proof of global simplicity. No new end-to-end Lean proof or explicitly emitted object-language ZFC derivation is claimed.

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1 The result and the meaning of simplicity

The task is to give a fixed polynomial whose lack of integer solutions expresses syntactic consistency of ZFC, with a construction simpler than the existing alternatives. There is no agreed total ordering of polynomials by simplicity. Number of variables, degree, number of terms, coefficient height, description length, and complexity of a correctness proof can favor different constructions. A claim of the globally "simplest" equation would need both a specified ordering and an optimality argument. We make neither claim.

This supplement improves a particular useful region of that comparison: *explicit quartics with ordinary literal coefficients, obtained directly from a finite proof calculus.* The starting equation is the 8,619-integer companion of the 3,609-natural-variable shared-CRT prime-lift construction at repository commit `278bc68df1af125d62f9cbfe28c9fd3ebcf50d5b` [1, 2]. Its arithmetic represents C222 derivations and uses neither the K215 Turing machine nor Jones's universal Diophantine expression.

Theorem 1.1 (The explicit new equation)**.** *Let $P \in \mathbb{Z}[X_0, \ldots, X_{3775}]$ be the polynomial in the delivered* ***equations/main/expanded_polynomial.json****, equivalently the sum of the squared residuals in the adjacent* ***equation.json****. Then*

$$\boxed{\mathsf{ZFC} \vdash \left[\neg\exists \boldsymbol{x} \in \mathbb{Z}^{3776}\ P(\boldsymbol{x}) = 0\right] \iff \mathrm{Con}(\mathsf{ZFC}).}$$

Here $\mathrm{Con}(\mathsf{ZFC})$ *means that no finite derivation of falsity exists from the repository's specified sealed ZFC axiom predicate. The ordinary proof uses the named C222/calculus and finite arithmetic representation theorems listed in Section 3, together with the new transformations proved below. The polynomial has the exact measurements*

$$\begin{aligned} \textit{unknowns} &= 3776, & \textit{degree} &= 4, \\ \textit{nonzero monomials} &= 20194, & \max|\textit{coefficient}| &= 47959. \end{aligned}$$

Every unknown ranges over all *integers. There are no free parameters or unspecified coefficient choices.*

The notation $\mathsf{ZFC} \vdash$ states ordinary mathematical provability. The proof consists of uniform finite constructions, elementary arithmetic, classical number theory, and the named ordinary logical calibration theorems. It does not assert that this archive contains a new machine-checked derivation of that sentence. In particular, the proof does not infer ZFC-provability merely from the existence of a Lean declaration. The distinction is discussed in Section 8.

1.1 Measured improvements

Two source presentations give different useful final equations. More aggressive elimination of the affine-source intermediate reaches 3,671 variables, but increases coefficient height substantially. Starting from the retained-wire presentation and limiting substitutions to two-term right-hand sides instead yields the recommended 3,776-variable polynomial. It has only 105 more variables, 6,077 fewer monomials, and approximately 79,972 times smaller coefficient height. It improves all tabulated size measures of the frozen 8,619-variable parent, with degree fixed at four. The 7,058-variable equation remains available when minimizing coefficient height itself is the priority.

Small-variable alternatives already exist. The repository has nine natural-variable and ten integer-variable equations of much higher degree; its latest accepted ten-integer table includes degree

Construction	Integer unknowns	Quadratic residuals	Quartic terms	Coefficient height
Frozen direct parent	8,619	4,619	39,811	127,285,349
Signed quotients	6,628	3,733	31,237	127,285,349
Reduced sign guards	4,879	3,150	22,910	127,285,349
Retained-wire guards	7,058	5,297	24,277	1,992
Fewer-variable companion	3,671	1,942	26,271	3,835,369,234
Recommended P	3,776	2,015	20,194	47,959

Table 1: Complete materialized polynomials, all over unrestricted integers and all of exact degree four. "Height" refers to the expanded polynomial, not just the residual coefficients.

1,080,888 at 512 arithmetic operations [5]. Sun's 2026 preprint gives a general ten-integer-variable representation theorem [6]. The present work does not improve that variable count. Likewise, proof-theoretic Diophantine encodings independent of a Turing-machine simulation are not a new general idea; Carl and Moroz developed such an approach explicitly [8]. Our contribution is the concrete reduction of a fully specified direct quartic and the reusable exact transformation certificates.

2 A literal polynomial, not an arithmetic shorthand

All construction variables are enumerated in a fixed list. In a sparse row, an entry $(c, [i_1, \dots, i_s])$ denotes the ordinary monomial $cX_{i_1} \cdots X_{i_s}$; the empty list denotes the constant c. Repeated indices denote ordinary integer powers. Every residual in the final equation has $s \le 2$.

Write the final residual list as $f_0, \dots, f_{2014}$. The defining equation is simply

$$P(\boldsymbol{X}) = \sum_{r=0}^{2014} f_r(\boldsymbol{X})^2 = 0, \qquad \boldsymbol{X} \in \mathbb{Z}^{3776}. \tag{1}$$

Both every f_r and every collected coefficient of P are provided. There are no variable exponents, factorials, binomial coefficients, bounded universal quantifiers, remainder operators, or residual computability predicates in these files. Such operations occur only in the proof explaining how the earlier natural parent was constructed.

Over the integers, a sum of squares vanishes exactly when each summand vanishes. Thus the residual presentation and the expanded polynomial have identical zero sets. This is the only conjunction operation in the final definition. The upper bound $\deg P \le 4$ follows immediately; the exact expansion contains nonzero degree-four terms, proving equality. The independent checker also verifies that every declared variable occurs in the expanded polynomial.

The JSON format is not an additional mathematical parameter. It is an unambiguous finite coefficient table, in the same sense as a printed list of integers. The source, construction algorithms, certificates, and hashes make the table reproducible. A human-readable residual listing is included for inspection without a JSON viewer.

3 The established direct source and its logical meaning

This section identifies the imported results and summarizes the unbounded source argument. It is included to prevent a local polynomial optimization from being mistaken for an independent construction of the entire consistency encoding.

3.1 The exact natural parent

Let $\mathcal{S}_0$ be the frozen natural system in `signed_quotients/source/natural_equation.json`. Its complete byte string has SHA-256

`e851ff1d420a39be069e63f6f117557356c38e6d8272a481258fe38cb38dc6ea.`

It has 3,609 nonnegative coordinates and 2,949 residuals of degree at most two. Its squared sum has 15,752 monomials and height 127,285,349. Of its coordinates, 1,670 are semantic inputs and 1,939 are retained arithmetic wires. The parent generator is bundled and reconstructs the complete equation byte for byte. Its arithmetic calls expand 55 positive power graphs, 16 binomial graphs, and two factorial graphs into ordinary quadratic rows.

The same source also has a retained-wire natural presentation with 6,180 variables, 5,520 quadratic rows, 17,421 quartic monomials, and coefficient height 1,992. Eliminating 2,571 nonnegative affine wire definitions yields exactly $\mathcal{S}_0$. Each deleted wire has a unique nonnegative polynomial restoration, so the two natural presentations have equivalent zero existence. Both complete parents are regenerated by the supplied source verifier. The retained presentation, whose literal file hash begins `39b3e264c051`, is the starting point of the recommended low-height final equation.

The established ordinary source theorem is

$$\exists \boldsymbol{y} \in \mathbb{N}^{3609}\ \mathcal{S}_0(\boldsymbol{y}) \quad \longleftrightarrow \quad \text{there exists a finite syntactic ZFC refutation.} \tag{2}$$

We import the proved source and representation theorems behind this statement, rather than assume a new simulator calibration. The current repository distinguishes this direct chain from some native machine/coefficient chains whose complete formal bridges remain separate research tasks [3, 4].

3.2 Numerical calculus

The numerical calculus C222 has truth label 0, falsity label 1, a fixed finite set of numerical rule families, modus ponens, and generalization. Its constructors are ordinary polynomials derived from Cantor pairing; for example,

$$\mathrm{Imp}(a, b) = 2(a + b)(a + b + 1) + 4b.$$

The relation, quantifier, and equality constructors use fixed offsets. The supplied direct source includes all twelve padded single-conclusion branches. Unused premise positions receive the truth label 0.

The decisive source declarations are pinned to commit `9a33342a03d1` (the full revision is in the source ledger):

```
KEndpoint.c222_iff_zf_refutation :
  KCode222.C222 Code254.F <-> BProv ZFax_s [] Form.fBot

ConstructibleChoice.zf_refutation_iff_zfc_refutation :
  BProv ZFax_s [] Form.fBot <-> BProv ZFCax_s [] Form.fBot
```

These are unconditional calculus/refutation interfaces, not hypotheses about execution of a Turing machine. The ordinary arguments use first-order soundness and completeness, finite proof transformations, and the constructible-model relative-consistency theorem. They do not require an externally transitive model or an assumption of Con(ZFC). We retain the source pins and attribution and do not claim to have rebuilt the complete Lean dependency closure.

3.3 Finite flow and the product carrier

For finitely many valid rule instances let c_i be the conclusion and p_i, q_i their truth-padded premises. The balance at every positive label j is

$$\#\{i : c_i = j\} = [j = 1] + \#\{i : p_i = j\} + \#\{i : q_i = j\}. \tag{3}$$

A proof tree gives these balances. Conversely, take the closure of truth under the finitely supported rules. If 1 were outside this closure, sum the conclusion-minus-premise balance over labels outside it. Every rule whose conclusion is outside has at least one premise outside, so its contribution is nonpositive. Rules whose conclusion is inside also contribute at most zero. The target contributes 1, a contradiction. This argument allows cycles, repeated premises, and repeated rule instances. It uses one conclusion per rule and nonnegative multiplicities.

If there are $t \geq 1$ instances with labels at most $m \geq 1$, define

$$A(Z) = \prod_{i<t}(1 + c_i Z), \qquad B(Z) = (1 + Z)\prod_{i<t}(1 + p_i Z)(1 + q_i Z), \qquad M = (m + 1)^t.$$

Equality $A(M) = B(M)$ first bounds the number of nonconstant factors of B by t: otherwise $B(M) > M^{t+1}$, whereas $A(M) < M^{t+1}$. Consequently $A(1), B(1) \leq M$, and each nonconstant coefficient is strictly less than M, because the constant coefficient is one. A least nonzero coefficient of $A - B$, reduced modulo M, would be both divisible by M and smaller than M in absolute value. Thus $A = B$ as polynomials. Multiplicities of the distinct roots $-1/j$ give (3).

A single accumulator encodes the equality by positive transitions

$$S_0 = M + 1, \qquad S_t = 1, \qquad S_{i+1}(1 + c_i M) = S_i(1 + p_i M)(1 + q_i M).$$

A proof tree supplies integral accumulator values by following its open obligations. The accumulator sequence is beta-coded with moduli $1 + (i + 1)y$. The proof, label bound, carrier, trace, beta code, and later row-witness bound are chosen in that order. The label bound and the bound on all arithmetic witnesses are distinct, so there is no circular size requirement.

3.4 Uniform CRT and prime-power lifting

The bounded row relation has nonnegative-coefficient polynomial sides E, F, an index bound N, external parameters z, and coordinate bounds $u_1, \ldots, u_d$. Put

$$H = E(N, z, u) + F(N, z, u) + N, \quad T = H!, \quad D_i = 1 + (i + 1)T, \quad B = \prod_{i<N} D_i.$$

The natural source uses encoded columns A_j and encoded index C satisfying

$$B \mid \binom{A_j}{u_j + 1} \qquad (1 \leq j \leq d), \tag{4}$$

$$B \mid 1 + (C + 1)T, \qquad E(C, z, A) \equiv F(C, z, A) \pmod{B}. \tag{5}$$

These conditions are equivalent to the existence of a bounded valid row for every $i < N$ [2].

For soundness choose a prime $p \mid D_i$. It is coprime to $H!$, hence $p > H$. Multiplying (4) by $(u_j + 1)!$ and using the falling-factorial identity shows that p divides some $A_j - r$ with $0 \leq r \leq u_j$. This remains true when $A_j \leq u_j$, since then the falling product has a zero factor. Therefore each column has a bounded decoded representative modulo p. The index congruence yields $C = i$ modulo p. Both decoded values E, F are at most $H < p$, so their congruence is equality.

For completeness the D_i are pairwise coprime. For each coordinate let $R_j = (u_j + 1)!$ and let R_{ij} contain precisely the prime-power part of R_j supported on primes dividing D_i. Choose the column by CRT with

$$A_j \equiv w_j^{(i)} \pmod{D_i R_{ij}}, \qquad C \equiv i \pmod{D_i}.$$

These lifted column moduli remain pairwise coprime. The falling product is divisible by $D_i R_{ij}$; cancel R_{ij} and then the coprime remaining factor R_j/R_{ij} to obtain (4). A positive column is chosen modulo the product of the lifted periods. Merely adding B is not sufficient without factorial coprimality: for instance, $3 \mid \binom{1}{3}$ but $3 \nmid \binom{4}{3}$.

This construction does not require $H \geq u_j + 1$. All factorization and CRT operations here construct witnesses in the proof; they are not operators left in the final polynomial. The supplied finite Pell-based power, binomial, and factorial representations replace the displayed arithmetic macros by ordinary quadratic rows. Their exact call-site positivity hypotheses and coefficient identities are audited by the bundled original independent checkers.

The central arithmetic interfaces are `PellRepresentation.criterion_iff_pell` and `PowerRepresentation.criterion_iff_pow`. Their source uses the finite Matiyasevich Pell criterion and its power consequence, respectively; selected exact source files are retained under `source_calibration`. Appendix A displays the particular positive power criterion used here. This is a named number-theoretic dependency, not an unexpanded power operation in the final coefficient table.

4 Removing the natural encoding of signed quotients

The first new reduction changes the domain of a witness to the domain the mathematical relation actually needs. An integer congruence quotient may be negative. The natural parent represents it by a difference of two nonnegative witnesses, then computes the two products separately. Applying a generic natural-to-integer conversion to those witnesses adds more unnecessary variables.

Lemma 4.1 (Private signed quotient)**.** *Suppose a natural system contains private coordinates q_+, q_-, u, v used only in*

$$u = Mq_+, \qquad v = Mq_-, \qquad E + c(u - v) = 0,$$

where M is a nonnegative expression in retained natural coordinates, $c \in \mathbb{Z}$, and E is independent of these four coordinates. Replace the three rows by

$$E + cMz = 0, \qquad z \in \mathbb{Z}.$$

Existence of a solution of the original natural system is equivalent to existence of a solution of the new mixed-domain system.

Proof. Projection takes $z = q_+ - q_-$. Conversely, from a signed witness z choose

$$q_+ = \max(z, 0), \quad q_- = \max(-z, 0), \quad u = Mq_+, \quad v = Mq_-.$$

These are natural numbers and satisfy all three original rows. Private use ensures that no other constraint has changed. The proof also covers $M = 0$ and does not divide by the modulus. □

The replacement is not a bijection: many pairs (q_+, q_-) have the same difference. This is immaterial for existential equivalence, and the displayed canonical lift is sufficient. It would be

incorrect to claim that simply allowing both old quotient coordinates to be signed and discarding their guards leaves an arbitrary natural source sound; the private-use property and the explicit lift are essential.

In the frozen parent there are exactly 221 qualifying pairs. The certificate checks the complete occurrence set of each pair, its common affine modulus, its two private product wires, and their opposite linear coefficients in the single final use row. Two additional private affine bridge wires are first substituted out. Each quotient block then removes three coordinates and two defining rows. Thus

$$3609 - 3 \cdot 221 - 2 = 2944, \qquad 2949 - 2 \cdot 221 - 2 = 2505.$$

The resulting core has 221 signed quotient coordinates, 1,228 natural semantic coordinates, and 1,495 natural arithmetic wires.

All retained gate definitions have nonnegative coefficients in retained natural coordinates and contain no signed quotient. The common moduli also have this property. Therefore the restoration in Lemma 4.1 genuinely returns natural product wires; this is verified syntactically rather than assumed from variable names.

4.1 The initial integer lift

For completeness, the basic nonnegativity certificate is

$$x = a^2 + b^2 + c(c+1), \qquad a, b, c \in \mathbb{Z}. \tag{6}$$

Lemma 4.2. *An integer x admits (6) if and only if $x \geq 0$.*

Proof. Every summand is nonnegative for integer arguments. Conversely, $4x + 1$ is not of the form $4^r(8s + 7)$, so Legendre's three-square theorem applies. Exactly one square is odd modulo four. Write the roots as $2a, 2b, 2c + 1$, subtract one, and divide by four. □

Initially guard all 1,228 remaining semantic natural coordinates using (6). The arithmetic wires are natural by topological induction. This gives

$$2944 + 3 \cdot 1228 = 6628 \quad \text{variables}, \qquad 2505 + 1228 = 3733 \quad \text{rows}.$$

The next two sections show why most of those guards are unnecessary.

5 Normalizing signs by exact symmetries

A natural witness may be needed only up to a sign that the equations cannot distinguish. In that case it is enough that every integer solution has an equivalent representative with the desired sign. This is a different reason for omitting a guard than proving that the original coordinate is automatically nonnegative.

5.1 The general sign criterion

For a quadratic residual $f(\boldsymbol{X}) = \sum_\alpha c_\alpha \boldsymbol{X}^\alpha$, replace X_i by $(-1)^{s_i} X_i$, where $s \in \mathbb{F}_2^n$. The residual changes only by an overall sign if

$$\langle \alpha - \beta, s \rangle = 0 \pmod 2 \quad \text{for every pair of monomials in } \operatorname{supp}(f). \tag{7}$$

The coefficients need not be positive. Condition (7) is an exact finite linear system over $\mathbb{F}_2$, and it guarantees $f((-1)^s X) = \pm f(X)$ for every integer assignment. Its intersection over all residuals is a diagonal sign-symmetry group of the entire zero set.

Lemma 5.1 (Independent sign normalization)**.** *Suppose the core system has sign symmetries $\sigma_1, \ldots, \sigma_k$ such that σ_i flips a selected semantic coordinate z_i and changes no other semantic coordinate. Suppose none of the z_i is guarded. Then every solution of the core plus the remaining semantic guards can be changed to one with $z_1, \ldots, z_k \geq 0$, while preserving all those guards and all core zero equations.*

Proof. Apply σ_i exactly when $z_i < 0$. It changes z_i to its absolute value and leaves every other semantic coordinate unchanged. Diagonal sign changes commute, so later choices do not undo earlier normalizations. Every core residual is multiplied by a unit ± 1, and each guarded semantic coordinate keeps its value. A zero coordinate needs no change. □

5.2 The 110 normalizations in the actual equation

The core parity matrix has rank 2,834 on 2,944 coordinates. The emitted certificate supplies 110 independent sign symmetries, two for each of the 55 power graphs. The checker does not trust a rank calculation as a semantic premise: it directly checks (7) for every generator and every actual residual.

The first symmetry negates the coordinate Z in the power certificate and its product wire WZ. The product definition is multiplied by -1, while occurrences of its square are unchanged. The second negates the Pell coordinate U and the two signed congruence quotients attached to modulus U. Products Uq are unchanged, and the Pell equation uses U^2. All other semantic coordinates are fixed.

The first case illustrates why inspecting only the exponents of Z in the final core is insufficient: Z occurs linearly in the definition of WZ. Coordinated sign changes of a semantic witness and its wires reveal the actual symmetry. The second case is enabled by the earlier signed-quotient reduction; natural quotient pairs would obscure the simple sign action.

These 110 normalizations save 330 guard auxiliaries. Some arithmetic wires may be negative before normalization, so naturality of all wires must be proved *after* normalization, as in the next section.

6 Recovering signs from the equations themselves

The remaining guard optimization is an exact proof of lower bounds, not an assumption that an arithmetic-looking variable is positive. The proof certificate starts with a chosen seed set of guarded semantic coordinates and the normalized coordinates from Lemma 5.1. It then derives lower bounds for every natural core coordinate.

6.1 A finite positivity calculus

Maintain a proved lower bound $L_i \in \{0, 1, 2\}$ for some coordinates; unbounded coordinates may still have either sign. For a polynomial G, substitute $X_i = L_i + T_i$ for the bounded coordinates, with $T_i \geq 0$. If the resulting coefficients are nonnegative and every unbounded variable has even exponent in each monomial, then $G \geq 0$. If the constant coefficient is positive, then $G > 0$.

This criterion is deliberately sufficient rather than complete. It can be checked by finite integer polynomial arithmetic. Squares of unbounded coordinates are allowed, which is useful before every wire has acquired a sign bound.

Lemma 6.1 (Positive-multiplier inference)**.** *Suppose an actual residual, oriented by a possible factor* -1*, has the form*

$$A(\boldsymbol{X})x = B(\boldsymbol{X}),$$

where A, B *do not contain* x*, and previously proved bounds establish* $A > 0$*. Then:*

1. $B \geq 0$ *implies* $x \geq 0$*;*
2. $B > 0$ *implies* $x \geq 1$ *for integer* x*;*
3. $B - A > 0$ *implies* $x \geq 2$ *for integer* x*.*

Proof. The first two conclusions follow from multiplication by a positive number. For the third, $A(x-1) = B - A > 0$, so $x > 1$; integrality gives $x \geq 2$. No division is inserted in the polynomial system. □

Examples include $A = 2 + s$, which makes the guard on A redundant when $s \geq 0$, and the less obvious row

$$B = 1 + 4Yq_4.$$

Once $B \geq 2$ and $Y \geq 1$ are known, it forces $q_4 \geq 1$ without a guard on q_4. This is a genuine use of a positive multiplier, not mere propagation of nonnegative coefficients.

6.2 The certified seed set

The complete certificate keeps 645 semantic guards. Its 110 sign normalizations and 2,382 ordered lower-bound inferences recover nonnegativity of all 2,723 natural core coordinates. The 221 signed quotients retain no sign bound. Among the 1,228 semantic natural coordinates, the exact accounting is

$$1228 = 645 \text{ guarded } + \ 110 \text{ normalized } + \ 473 \text{ inferred.}$$

Some coordinates receive stronger bounds after an initial bound, which explains why the number of inference steps exceeds the number of new coordinates proved nonnegative.

The seed selection is an optimization heuristic. Its success is not a mathematical hypothesis: a separate checker verifies the supplied ordered proof, including the strict positivity of every multiplier and the exact polynomial row used at every step. It does not call the optimizer. Minimality of the 645-coordinate seed set is not asserted.

Theorem 6.2 (The 4,879-variable equation)**.** *The system consisting of the 2,505-row signed core and the 645 displayed three-square guards has an integer solution if and only if the frozen natural parent has a natural solution. Its squared sum has 4,879 integer coordinates, 3,150 quadratic residuals, 22,910 nonzero terms, and coefficient height 127,285,349.*

Proof. For soundness first normalize the 110 signs. The retained guards remain valid. Replay the lower-bound proof to recover all natural core coordinates. Apply the signed-quotient lift and the two affine bridge restorations to obtain a natural parent solution. For completeness, project any natural parent solution to the signed core. Every one of its 645 guarded coordinates is natural, so the three-square theorem provides the fresh guard auxiliaries. All other omitted guards are unneeded in this direction.

The variable and row counts are $2944 + 3 \cdot 645 = 4879$ and $2505 + 645 = 3150$. Exact expansion of the supplied finite rows gives the remaining measurements. □

Corollary 6.3 (The retained-wire companion)**.** *The same methods applied to the retained-wire natural parent give an equivalent integer equation in 7,058 variables, with 5,297 quadratic rows, 24,277 monomials, and coefficient height 1,992.*

Proof. The exact signed-quotient certificate first removes 442 private affine bridges and the same 221 quotient pairs. The resulting core has $6180 - 442 - 3 \cdot 221 = 5075$ variables and $5520 - 442 - 2 \cdot 221 = 4636$ rows. It retains 1,228 semantic natural inputs, 221 signed quotients, and 3,626 arithmetic wires.

The sign certificate uses the same 110 independent semantic normalizations, now with the additional literal wire sign changes. It retains 661 guards, 16 more than the affine-source case, and recovers 457 other semantic signs. Its 4,942 ordered bound steps prove nonnegativity of all 4,854 natural core coordinates. The same normalization, inference, and quotient-restoration arguments apply. Thus the integer dimensions are $5075 + 3 \cdot 661 = 7058$ and $4636 + 661 = 5297$. Full expansion gives the remaining exact measurements. □

7 Polynomial elimination without a degree increase

After sign normalization and guard reduction, every actual coordinate of the displayed system ranges over $\mathbb{Z}$. This licenses an especially simple second reduction: solve monic equations and substitute them. There is no need to preserve a separate natural domain at this stage, because its required effects are already encoded by the remaining rows.

Theorem 7.1 (Degree-preserving monic projection)**.** *Consider a system of polynomials of degree at most two over $\mathbb{Z}$ with one distinguished equation*

$$z - R(\boldsymbol{x}) = 0,$$

where R is independent of z. Assume either:

1. *R has degree at most one; or*
2. *R has degree at most two, and every other residual contains z only as a scalar-linear term cz.*

Delete the distinguished row and substitute R for z everywhere else. The resulting residuals have degree at most two, and projection off z is a bijection between the two integer zero sets. Its inverse is the polynomial map $\boldsymbol{x} \mapsto (\boldsymbol{x}, R(\boldsymbol{x}))$.

Proof. Every source solution has $z = R(\boldsymbol{x})$ and therefore projects to a target solution. Conversely, adjoining that value makes the distinguished row and all substituted rows zero, uniquely. An affine substitution in a quadratic polynomial remains quadratic. In the second case each other row is $cz + G(\boldsymbol{x})$ and becomes $cR + G$, also quadratic. All coefficients remain integers, so there is no divisibility side condition or exceptional denominator. □

Remark 7.2. The second criterion excludes occurrences such as z^2 and zw. Substituting a quadratic expression there could raise the residual degree to four and the squared-sum degree to eight. The implementation checks actual monomials in every affected row, not just a textual occurrence count. Solving $2z = R$ without a divisibility certificate is also excluded; the pivot coefficient must be 1 or -1.

The algorithm repeatedly chooses a permitted sparse substitution. It records the exact current defining row, pivot variable, right-hand side, and full set of affected rows. Original variable and row

indices remain fixed throughout the certificate; surviving coordinates are renumbered only at the end. A complete reverse lift is obtained by processing the certificate backwards. Composition of the polynomial lifts gives an integer polynomial inverse to the final projection on the zero sets.

The recommended final pass starts with the 7,058-variable equation of Corollary 6.3 and removes 3282 variables: 1915 by affine substitution and 1367 by the linear-occurrence quadratic rule. It permits at most two monomials in each substituted right-hand side, as a size heuristic to avoid dense coefficient growth. The support caps are not correctness assumptions. A separate checker imports no optimizer and replays every exact sparse substitution, including changes to guards.

For the smaller-variable companion, a second pass starts with the 4,879-variable affine-source equation and removes 1,208 coordinates: 461 affine and 747 quadratic substitutions. This gives 3,671 variables, 1,942 residuals, and the larger coefficient height in the comparison table. That pass uses support caps 18 and 24, respectively. Both complete equations and both exact inverse-map certificates are retained.

Applying Theorem 7.1 successively to Corollary 6.3, then using (2), proves Theorem 1.1. The same argument applied to Theorem 6.2 proves the consistency equivalence for the 3,671-variable companion. The coefficient differences recorded in Table 1 are real; a final polynomial is not identical to its parent polynomial. What is preserved at this stage is a polynomial bijection of integer zero sets.

8 Why the equivalence is provable in ZFC

There are three distinct assertions that must not be conflated:

1. a standard-model equivalence checked separately for selected finite inputs;
2. an ordinary uniform mathematical proof available within ZFC;
3. a new emitted object-language proof or a complete kernel-checked implementation of that proof.

The delivered result is of the second kind, supported by exact coefficient and transformation certificates. It does not claim the third kind.

The new transformations are elementary and uniform. A quotient lift uses integer order and maximum; a sign normalization uses finitely many polynomial symmetries; the lower-bound proof is a finite sequence of ordered-ring implications; and each monic elimination is an exact polynomial identity with a polynomial inverse. ZFC proves the required laws of its internally defined integers and proves the three-square theorem. Therefore the finite certificates can be expanded into ordinary ZFC derivations of these transformations.

The source arithmetic also uses uniform theorems: finite products, induction, prime factorization, CRT, Pell identities, and syntactic proof transformations. The carrier and CRT arguments apply to every internal natural number in a model of ZFC, including numbers that are nonstandard from an external viewpoint. They are not an appeal to the assertion that a particular finite diagnostic test happened to pass. The ordinary C222 calibration and ZF/ZFC refutation transport are named logical theorems with set-theoretic proofs; their use is explicit in the dependency ledger.

The consistency sentence is syntactic. The theorem states

$$\mathsf{ZFC} \vdash (\exists \boldsymbol{x} \in \mathbb{Z}^{3776}\ P(\boldsymbol{x}) = 0) \iff (\exists p\ \mathrm{Proof}_{\mathsf{ZFC}}(p, \bot)).$$

Negating both sides yields the displayed no-zero form. No soundness or consistency assumption is used to prove the biconditional itself. In particular, this work does not prove that P lacks solutions.

If ZFC is consistent, a ZFC proof that P lacks solutions would, through this biconditional, give a ZFC proof of its own consistency. The construction is compatible with the second incompleteness theorem.

The archive includes ordinary dependency proof notes and pinned source interfaces, but not a newly rebuilt Lean checkout or a line-by-line formal ZFC proof. That remaining formalization task should be described accurately. It is different from leaving an unidentified coefficient, an arbitrary compiler choice, or a conjectural simulation premise in the polynomial.

9 Exact verification and reproducibility

The verification is designed around the risks introduced by the new construction: incorrect occurrence restrictions, sign changes that affect another guarded variable, circular positivity arguments, unsafe polynomial substitutions, and a mismatch between the stated residuals and their expanded squared sum.

The release verifier checks the following distinct objects.

1. **The fixed source.** The pinned direct arithmetic generator reproduces the complete natural parent. Original independent auditors check the actual twelve calculus branches, positive-power interfaces, and affine restoration. The package contains the relevant mathematical proof notes and licenses.

2. **Signed quotient blocks.** Every quotient and product use is located in the complete source row list. The exact common modulus, opposite signs, private-use restrictions, and nonnegative restoration of all remaining arithmetic wires are checked.

3. **Sign symmetries.** Every monomial of every core residual satisfies the parity test for each supplied generator. The checker also verifies that only the intended semantic coordinate changes.

4. **Lower-bound derivations.** Each step cites an actual row, uses only earlier bounds, and establishes strict positivity of its multiplier by exact shifted polynomial arithmetic. At the end, every intended natural coordinate has a nonnegative bound.

5. **Elimination.** A separate implementation reconstructs the solved equation and substitutes it into every actual occurrence. It verifies the degree criterion, deleted rows, surviving indices, and exact final renumbering.

6. **The polynomial.** All squared residuals are fully expanded and collected. Every coefficient, monomial, degree, variable, term count, and height is compared to the released file. Hashes identify the exact data subjected to these checks.

These checks use exact integer arithmetic. They do not rely on floating-point agreement, randomly selected zeros, or a finite search for a contradiction. The checkers use explicit failure conditions; Python's `-O` option does not disable their acceptance logic. Normal and optimized runs are included in the release evidence. The independent checkers do not import the new optimization routines.

The universal proofs establish why a syntactically valid certificate preserves existence for all witnesses. The finite checker establishes that the delivered coefficients and certificates instantiate those theorems. Neither role replaces the other.

9.1 Practical commands

Python 3.11 or later and its standard library suffice for the exact arithmetic verification. No network call or package installation is performed by the verification driver.

```
python3 verify.py
python3 -O verify.py
```

The README gives the additional regeneration commands, including the slower optimization pass for both supplied final endpoints. A standard TeX Live or MiKTeX installation builds this article:

```
cd report
pdflatex -interaction=nonstopmode -halt-on-error report.tex
pdflatex -interaction=nonstopmode -halt-on-error report.tex
```

The complete frozen data allow verification independently of later changes to ProveIt. The intended integration is a new candidate or incoming research supplement; no existing repository file needs to be overwritten.

10 What is new, and what has not been established

The specific new contribution is the combination of signed-domain compilation, exact sign symmetries, certified order propagation, and degree-preserving polynomial elimination on this fixed direct consistency equation. The first two methods interact: replacing natural quotient pairs by signed quotients exposes the Pell-modulus sign symmetry and makes its normalizing map particularly simple. The sign proof then removes guard variables; unrestricted polynomial elimination can finally remove semantic slack coordinates as well as arithmetic wires without maintaining an artificial natural domain.

The constituent mathematical ideas have recognizable precedents: differences of naturals, sign symmetries, three-square representations, and monic substitutions are classical. We do not claim that these general ideas were previously unknown. The exact combined application, finite certificates, and resulting full polynomial are the advance relative to the inspected repository snapshot. Establishing priority in the broader literature would require a separate review.

The following statements are deliberately not made:

- that this is the fewest possible number of unknowns;
- that degree four is minimal for this particular consistency encoding;
- that the guarded seed set or elimination order is optimal;
- that a literal polynomial with fewer variables must have a simpler proof or a shorter description;
- that the finite audits decide ZFC consistency; or
- that a new complete Lean proof or an explicit formal ZFC proof file is present.

The useful claim is narrower and fully measured: a large reduction of the explicit direct quartic's dimension, with the same logical target, the same exact degree, reproducible ordinary coefficients, and constructive proof maps.

11 Further research questions

1. Optimize guard seeds with a proved lower bound. The current seed set is found by a heuristic and certified afterward. The inference rules define a directed dependency structure with alternative prerequisites. Can an integer-programming or satisfiability formulation find a smaller seed set, and can a matching lower-bound certificate prove optimality within a clearly stated inference calculus? An improvement here saves three variables per deleted guard before subsequent eliminations.

2. Use larger sign groups or affine symmetries. The current certificate uses independent diagonal sign normalizations. More general transformations, such as a sign change accompanied by an integral translation of congruence quotients, may preserve the zero set even when diagonal parity alone does not. Such a transformation needs an explicit inverse and must preserve the necessary semantic signs. Classifying these symmetries of the positive power graph is a concrete next target.

3. Compile the arithmetic graph directly with mixed domains. Here the signed quotient reduction is applied to a frozen natural parent so that the comparison is exact. A compiler that tracks natural, strictly positive, sign-normalizable, and unrestricted coordinates from the outset could avoid creating redundant witnesses. Its output should still carry a simple independently checked certificate, rather than require trusting a more elaborate optimizer.

4. Jointly optimize dimension, sparsity, and coefficient height. The final monic elimination increases coefficient height. Different pivot choices or retaining selected scalar wires could yield a better dimension/height tradeoff. A useful search should maintain the complete expanded polynomial at each accepted endpoint and report a small nondominated set rather than promote one arbitrary weighted score.

5. Replace a complete arithmetic primitive. The direct construction contains 55 power graphs. A smaller fully proved positive power representation, or a representation shared across compatible calls, could yield a larger improvement than further local row simplification. The exact source/Pell index hypotheses must be retained. Replacing the graph by a plausible rational Pell condition without proving integrality is insufficient.

6. Produce the end-to-end internal proof artifact. The ordinary ZFC proof is assembled here, but its actual emitted internal derivation is still a separate deliverable. The new algebraic and sign certificates are finite and suitable for a small trusted checker. Formalizing these generic certificate validators once would make later size optimizations much less expensive to certify. The source calibration and the new literal coefficient binding should remain separate theorem interfaces.

7. Compare with small-variable encodings on description length. A quartic with thousands of moderate coefficients and a high-degree ten-variable polynomial optimize different resources. A fair comparison could measure a fixed binary description format, including all literal coefficients and any bounded coefficient recipes, together with the cost of a proof checker. This would make the request for a "simplest" equation more precise without confusing arithmetic operation counts with coefficient bit complexity.

A The arithmetic criterion behind the sign symmetries

For positive inputs n, k and natural output m, the parent power graph has 27 semantic auxiliary witnesses. Four difference-of-natural quotients become four signed witnesses after our first reduction, leaving 23 auxiliary coordinates before wire flattening. Write

$$\mathcal{N}(a, x, y) = x^2 - (a^2 - 1)y^2 - 1.$$

The full positive criterion consists of the following seventeen constraints. All auxiliary coordinates are initially natural except the four new congruence quotients z_1, z_2, z_3, z_4:

$$\begin{aligned}
A &= a_2 + 2, & Y &= k + y_k, & B &= b_2 + 2,\\
\mathcal{N}(A, X, Y) &= 0, & \mathcal{N}(A, U, V) &= 0, & \mathcal{N}(B, S, R) &= 0,\\
B &= 1 + 4Yq_4, & V &= Y^2(v_d + 1), & &\\
B + Uz_1 &= A, & S + Uz_2 &= X, & &\\
R + 4Yz_3 &= k, & X + Yn + Tz_4 &= YA + m, & &\\
2An &= T + n^2 + 1, & T &= m + g + 1, & &\\
W &= n + w_n, & W &= k + w_k, & &\\
\mathcal{N}(W + 1, A, WZ) &= 0. & & & &
\end{aligned}$$

These expressions are displayed before arithmetic wire flattening; their degrees are not the degree bound on the final residuals. The actual generator replaces their product expressions by the retained quadratic gate equations and shares repeated subexpressions.

The last norm rewrites as

$$A^2 = 1 + W^3(W + 2)Z^2.$$

Since $A \geq 2$ and $W \geq n \geq 1$, it forces $Z \neq 0$ and $A > W \geq n$. Therefore the natural truncated difference $A \dot{-} n$ in the inherited power theorem equals the ordinary integer difference $A - n$ in the rearranged congruence. Also $Y \geq k \geq 1$, so $V = Y^2(v_d + 1) > 0$ and the zero-index Pell branch is absent. These facts justify the positive specialization of the imported power criterion.

The sign symmetries are now visible at the unflattened level: $Z \mapsto -Z$ changes no norm, and $(U, z_1, z_2) \mapsto (-U, -z_1, -z_2)$ changes neither the norms nor the congruences. The literal flattened certificate checks every wire needed to realize the same action. Bounds inferred from these rows are used only after the necessary natural coordinates are guarded or normalized, so the proof does not silently assume $V > 0$ when $v_d = -1$ might otherwise be possible.

B Data and proof-certificate conventions

Variable lists are ordered and indexed from zero. Each sparse polynomial uses sorted monomials and nonzero integer coefficients; duplicates are collected. The final expanded table has exactly 20194 entries. The main equation and its expansion are identified by the hashes in `equations/main/BUILD.json` and the release manifest. The polynomial hash is

`28e218143296c666cd387ef4d9ccc5a4ab0d45506a59ef2fb946c37291a17337`

For an elimination step, the certificate stores an original coordinate index v, a current monic defining row, a polynomial expression R, and the full set of current rows containing v. The checker first verifies the identity of the defining row with $\pm(X_v - R)$, then checks the degree criterion and

substitutes into every occurrence. The certificate can be interpreted backwards without rerunning the optimization search.

For a sign step, the certificate stores the list of flipped coordinate indices and the unique semantic pivot. For a lower-bound step it stores the target coordinate, new bound, and the actual source row index. The multiplier and right-hand side are reconstructed from that row by the checker. Therefore a claimed lower bound cannot smuggle an additional equation into the proof.

C A small illustrative quotient block

This example is explanatory and does not replace the full C222 instance. Suppose a retained natural modulus is $M = 7$ and a row requires a difference $E = -21$. The old equations

$$u = 7q_+, \quad v = 7q_-, \quad -21 + u - v = 0$$

can be replaced by $-21 + 7z = 0$. The signed witness $z = 3$ restores as $q_+ = 3, q_- = 0, u = 21, v = 0$. Reversing the sign of E gives $z = -3$ and restores with the two natural quotient parts exchanged. If $M = 0$, both systems require only $E = 0$, and the canonical lift still works. This last case explains why avoiding a division by M is valuable.

For an illustrative elimination, the rows

$$z - x^2 - y^2 = 0, \qquad 3z + w - 7 = 0$$

reduce to $3x^2 + 3y^2 + w - 7 = 0$, with inverse $z = x^2 + y^2$ and no degree increase. If the second row instead contains z^2, this quadratic elimination is disallowed by the certificate rule.

D Provenance and scope of the release

The comparison snapshot is fixed even if the repository advances while this research is in progress. The archive retains the exact natural source, original arithmetic generator and auditors, relevant ordinary proof notes, source-calibration interfaces, licenses, the new complete integer equations, independent checkers, and the article source and compiled PDF. Intermediate experimental outputs that do not support a released endpoint are omitted.

No remote repository mutation is needed to use the deliverable. The package can be reviewed and placed under `Computability/DiophantineConsistency/candidates` or `docs/incoming` at the owner's discretion. The release manifest records the exact file bytes; reproducing a PDF on a different TeX installation may legitimately change PDF metadata while leaving its mathematical content unchanged.

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